

Alwin Tay Jing Xue

LLM Researcher

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PROFESSIONAL SUMMARY

LLM Research Assistant (Contract), Software Business Analyst (Full-time), Machine Learning Engineer (Part-time) + 3 Internships with **3 years of professional experience** across LLM fine-tuning (LoRA/QLoRA), RAG architectures, business/data analysis, and machine learning model development.

PROFESSIONAL EXPERIENCE

LLM Research Assistant (Contract: 1 Year)

December 2025 - Present

Swinburne University of Technology (Sarawak Campus), Sarawak, Malaysia

- Fine-tuned Qwen3-Coder-30B-A3B-Instruct, a Mixture-of-Experts model with 3B active parameters, using a GRPO verifiable-reward LoRA loop on a 48 GB RunPod A40 GPU.
- Engineered the training pipeline in Python and PyTorch, cutting training memory from roughly 112 GiB to 10.4 GiB through 4-bit NF4 quantization with rank-16 LoRA adapters.
- Served the model through vLLM at a measured 41 GB, evaluating bug-localisation on the live serving path at 0.817 with a 95 percent confidence interval across six runs.
- Diagnosed and corrected a scoring defect that had inflated a reported metric, catching an automated grader crediting the model for echoing its own prompt rather than solving the task.
- Built a streaming FastAPI backend with a Next.js and TypeScript client over PostgreSQL, integrating the served model into a production document reasoning application.

LLM Research Intern

June 2025 - December 2025

Swinburne University of Technology (Sarawak Campus), Sarawak, Malaysia

- Implemented a Low-Rank Adaptation bias-detection workflow orchestrated in LangChain with Ollama inference, benchmarking Llama 3 8B against Direct Preference Optimization baselines using Qdrant retrieval.
- Fine-tuned a LoRA bias-mitigation adapter on Llama 3 8B in PyTorch with 4-bit quantization, training rank-16 adapters over attention and MLP projections on a curated preference dataset.
- Engineered a multi-stage LangChain evaluation pipeline that detects bias, generates a refined response, and scores both against the original using a structured comparison framework.

Software Business Analyst (Full-time)

Nov 2023 - June 2024

Finology Sdn Bhd, Kuala Lumpur, Malaysia

- Analysed business requirements across the full Software Development Life Cycle for two fintech platforms, Aeon Bank, a digital loan platform, and AHAM Asset Management, a digital investment platform.
- Translated business needs into 40+ technical Jira specifications with edge cases and data validation rules, reducing developer rework 30 percent across both engagements.
- Executed System Integration Testing and User Acceptance Testing across staging and production, owning functional specification documents through client sign-off in a cross-functional Agile Scrum team.

AI/Machine Learning Intern

Oct 2023 - Nov 2023

Hexabyn Technologies Sdn. Bhd, Sarawak, Malaysia

- Trained an EfficientNet-B0 defect classifier to 93.4 percent accuracy on a curated annotated image dataset through an iterative PyTorch training pipeline.
- Developed a Django interface consuming YOLOv8 detections into a functional quality-inspection workflow.

Quality Control Engineer Intern (Software Developer/Data Analyst)

Mar 2023 - Jul 2023

LONGi Malaysia Sdn. Bhd, Sarawak, Malaysia

- Built an internal Certificate Automation System in VB.NET, MySQL, and a WinForms interface, cutting manual certificate-generation effort 85 percent and processing time 75 percent.
- Automated Python ETL pipelines across 22 real-time manufacturing datasets, preprocessing solar-cell wafer dimensions for product certification and time-series defect prediction.

- Reduced certificate failures over 90 percent by implementing data validation checks, working with the IT department and Quality Control Engineers to map their certification process.

TECHNICAL PROJECTS

SYNTHIEN AI (AI Chatbot) [\[website\]](#)

June 2025 - Present

- Allow users to upload documents for summarization, flashcard generation, and journal article highlights with 38,912-token reasoning capacity. (Python, TypeScript, React.js, Node.js)
- Implemented RAG pipeline using Qdrant vector database, integrating Llama 3.1 8B, and Qwen 2.5 7B for enhanced document retrieval accuracy.

Skin Disease Classification System [\[code\]](#)

April 2025 - June 2025

- Developed with EfficientNet-B0 and Hugging Face DeiT models to accurately detect 10 dermatological conditions (e.g, Melanoma, Eczema, etc) with 84-100% accuracy.
- Built the PyTorch pipeline in Python and implemented weighted sampling, data augmentation, and early stopping for skin condition classification.

TECHNICAL SKILLS

- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, QLoRA, LoRA, PEFT, EfficientNet, DeiT, YOLOv8
- **LLM & Retrieval:** RAG, reranking, embeddings, Qdrant, pgvector, LangChain, prompt engineering, vLLM, Ollama
- **Data & Big Data:** Python, SQL, PySpark, Apache Spark, ETL, pandas, NumPy, time series analysis
- **Backend & MLOps:** FastAPI, PostgreSQL, Redis, Docker, AWS, GitHub Actions CI/CD
- **Languages:** Python, R, Java, C++, SQL, TypeScript, JavaScript
- **Spoken Languages:** English, Chinese (Mandarin), Bahasa Melayu

EDUCATION

Swinburne University Of Technology

Bachelor of Computer Science (Major in Data Science)

September 2024 - July 2026

LICENSE & CERTIFICATION

- PyTorch for Deep Learning (DeepLearning.AI) [\[link\]](#)
- TensorFlow Developer (DeepLearning.AI) [\[link\]](#)
- AI Engineer for Developers (DataCamp) [\[link\]](#)
- AI Engineer for Data Scientists (DataCamp) [\[link\]](#)
- Data Scientist Associate (DataCamp) [\[link\]](#)
- Product Management Certification by General Assembly [\[link\]](#)
- Statistical Analysis System (SAS) Certificate [\[link\]](#)

AWARDS & HONORS

- **[National]** TechFrontier Explorer × SCENT Demo Day Competition – 2nd Place Award [\[link\]](#)
- **[National]** TechFrontier Explorer × SCENT Programme – Grants Award [\[link\]](#)

REFERENCES

Available on request.